

EEE 312 – Digital Signal Processing I Laboratory
Jan 2026 Level-3 Term-I Section C2
Final Project Demonstration

Audio-Visual Speaker Isolation

SUBMITTED BY – GROUP 4 (C2)



Redwan Nur Alam Shilad
2206177



Ahnaf Hasan Adeeb
2206178



S. M. Khairul Islam Rifat
2206179



Md. Shahriare Arefin Shuvo
2206180

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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Outline

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| 2 Introduction | 7 Communication |
| 3 Design | 8 Project Management & Cost |
| 4 Implementation | 9 Future Work |
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Who presents



Redwan Nur Alam Shilad

Speaker 1 · Slides 1–8 · Problem



Ahnaf Hasan Adeeb

Speaker 2 · Slides 9–14 · Design & demo



S. M. Khairul Islam Rifat

Speaker 3 · Slides 15–22 · Evaluation



Md. Shahriare Arefin Shuvo

Speaker 4 · Slides 23–31 · Wrap-up



1. Summary / Abstract

Click a face.

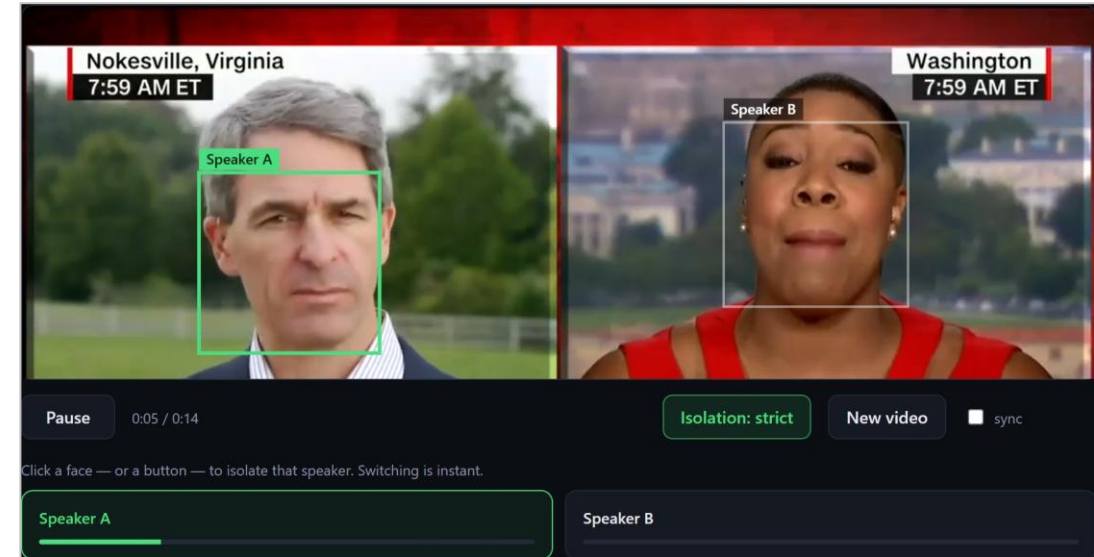
Hear only that voice.

Lips select the voice: a lip-conditioned network (AV-MossFormer2) extracts one voice per face

Our DSP removes the leftovers: refinement + coherent leak cancellation + exact-silence gate

Instant, click-free switching in the browser (Web Audio)

Generalises: any language, camera cuts, 2+ speakers



Our web app isolating Speaker A (best test video)

3.5 %

word error on our own
recording (audio-only: 110 %)

+0.78

lip ↔ voice correlation
(face A, isolated channel)

25 ms

click-free switch
between speakers



2. Introduction



Two of us reading different texts at the same time (our showcase recording)

1

One microphone, one waveform

$x(t) = s_1(t) + s_2(t)$: the voices are simply added

2

People solve it by watching lips

the “cocktail-party effect”

3

Our goal

pick a face → hear only that person

3.1 Design: Problem Formulation (PO(b))

3.1.1 Identification of Scope

In scope

- ✓ Videos with 2+ visible speakers
- ✓ One microphone: phone, TV, webcam
- ✓ Any language
- ✓ Process once, then play interactively

Out of scope

- ✗ Speakers never shown on screen
- ✗ Live real-time streams
- ✗ Faces smaller than ~150 px
- ✗ Music / non-speech sources



3.1 Design: Problem Formulation (PO(b))

3.1.2 Literature Review

Approach	Examples	Idea	Gap
Blind separation	SepFormer, MossFormer2	Splits voices by sound only	Doesn't know who is who; fails on similar voices
Separate + match	Our V2, lip–energy correlation	Pair voices to faces afterwards	Measured at chance (0.068 vs 0.068 on white noise)
Lip-conditioned (AV-TSE)	Looking to Listen '18, AV-MossFormer2 '24, Dolphin '26	Network is given the target's lips	Leaks when voices overlap continuously
Ours	AV-TSE + DSP + web player	Refinement, coherent leak cancel, re-ID	Interactive, measured on our own data



3.1 Design: Problem Formulation (PO(b))

3.1.3 Formulation of Problem

$$x(t) = \sum_k s_k(t) + n(t) \quad \text{given lips } V_k \rightarrow \text{estimate } \hat{s}_k(t) = f(x, V_k)$$

one mixed recording · one lip video per face · one clean voice per face out

1 Isolation

the other voice inaudible:
maximise signal-to-interference

2 Fidelity

the chosen voice unaltered:
no clipped words, low WER

3 Interaction

switch in ≤ 25 ms, exact silence
in pauses, audio-video in sync

Constraints: 8 GB laptop GPU · 16 kHz model · single microphone · Python 3.12



3.1 Design: Problem Formulation (PO(b))

3.1.4 Analysis

chance

Lip-loudness matcher scored 0.068 – the same as white noise (0.068)

→ **drop matching; condition on lips**

110 %

Audio-only separation interleaved both texts (word error 80–128 %)

→ **blind separation can't split similar voices**

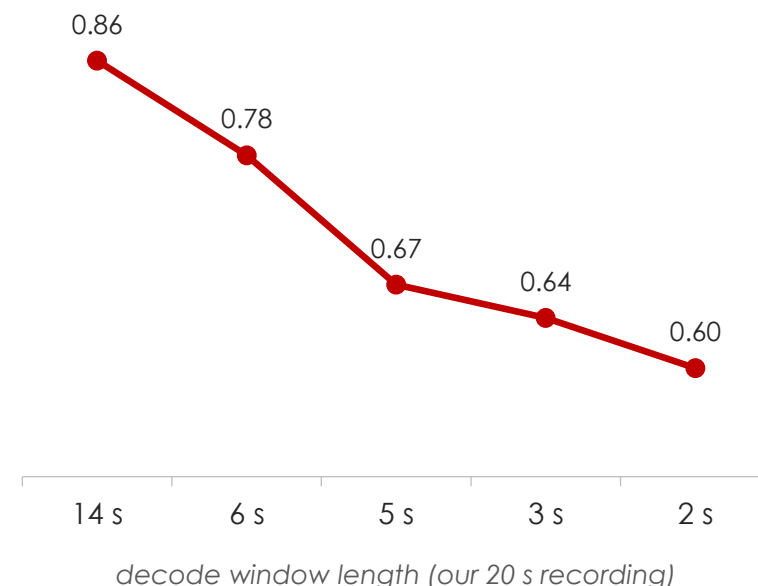
0.86 → 0.60

Channel similarity fell as decode windows shrank 14 s → 2 s

→ **run the model in its training regime**

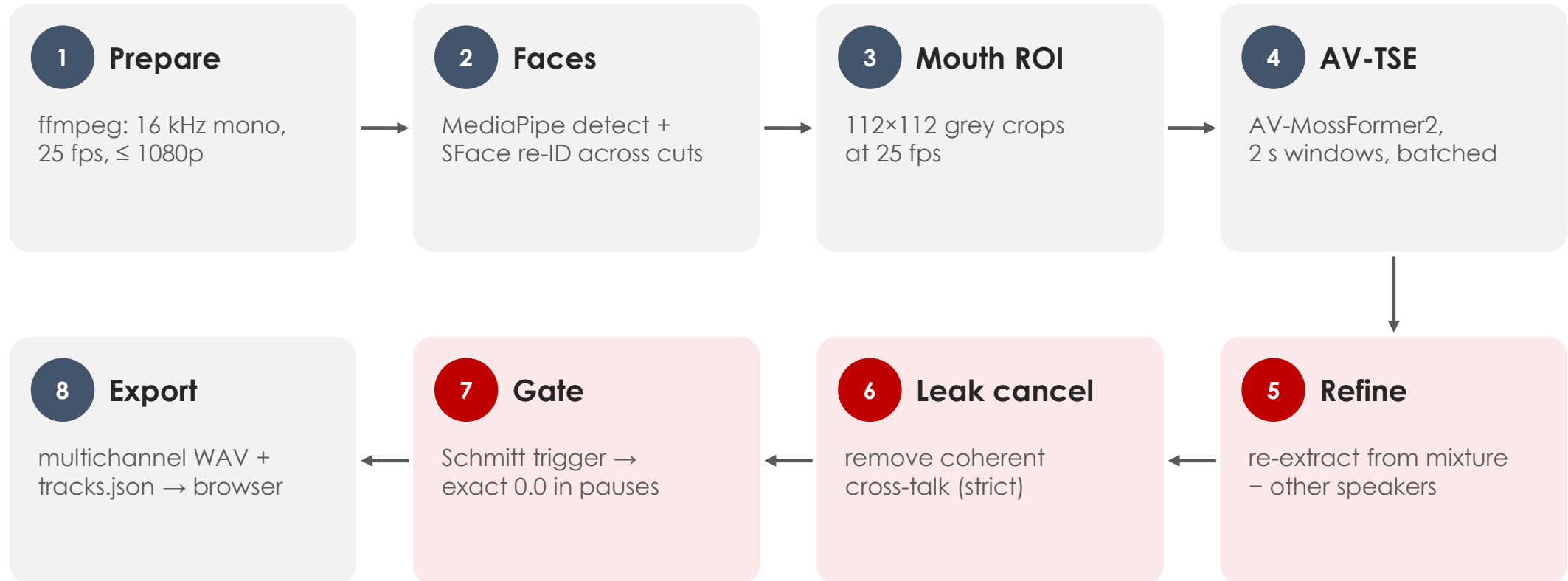
Channel similarity (lower = cleaner)

— Channel similarity (lower = cleaner)



3.2 Design Methods (PO(a))

System pipeline



■ our DSP stages ■ pretrained models / tools



3.2 Design Methods (PO(a))

The signal processing (PO(a))

1 Lip-conditioned extraction

$$\hat{s}_i = f(x, V_i)$$

2 s windows, centre 1 s kept, frame-aligned, raised-cosine seams

2 Refinement

$$x_i' = x - \sum_j (\hat{S}_j - \frac{1}{2} a_{ij} \hat{S}_i)$$

$\alpha = \langle \hat{S}_j, \hat{S}_i \rangle / \langle \hat{S}_i, \hat{S}_i \rangle$ (local STFT regression): never subtract your own voice

3 Coherent leak cancellation

$$\hat{S}_i \leftarrow \hat{S}_i - w \cdot a \cdot \hat{S}_j, \quad w = 1 / (1 + P_i/P_j)$$

only the part that is a copy of the other voice is removed

4 Silence gate

open -30 dB · close -40 dB · hold 120 ms

hysteresis + raised-cosine ramps $\rightarrow$ bit-exact 0.0, no clicks



3.2 Design Methods (PO(a))

Interactive player (Web Audio)

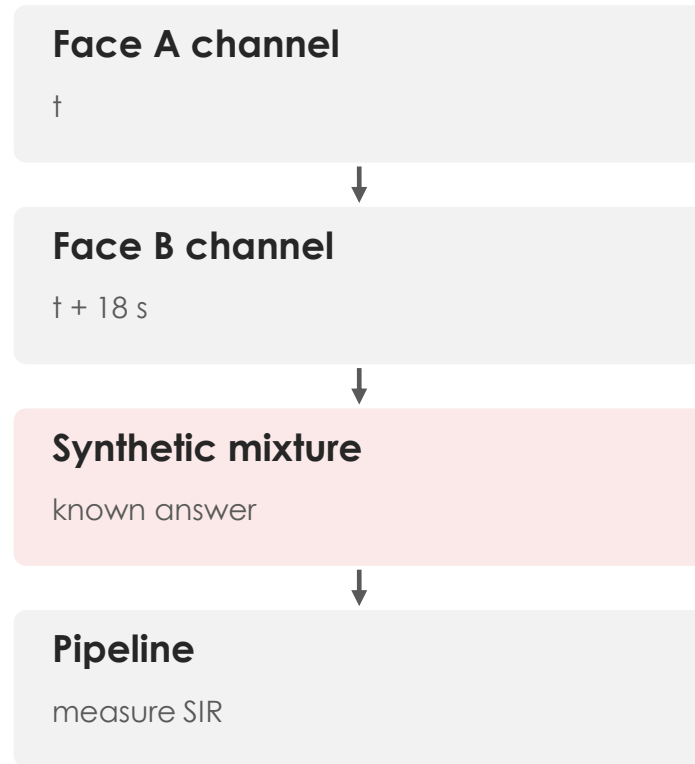


- ✓ **Click a face** 25 ms equal-power crossfade: $\cos^2\theta + \sin^2\theta = 1 \rightarrow$ no dip, no click
- ✓ **Isolation: strict $\rightleftharpoons$ natural** same crossfade, different channel offset $\rightarrow$ switch strength live
- ✓ **<video> muted, slaved to audio** `requestVideoFrameCallback`: $\pm 2\%$ `playbackRate`, re-seek if > 250 ms
- ✓ **Meters after the gain** a muted speaker reads a flat zero – the silence is visible

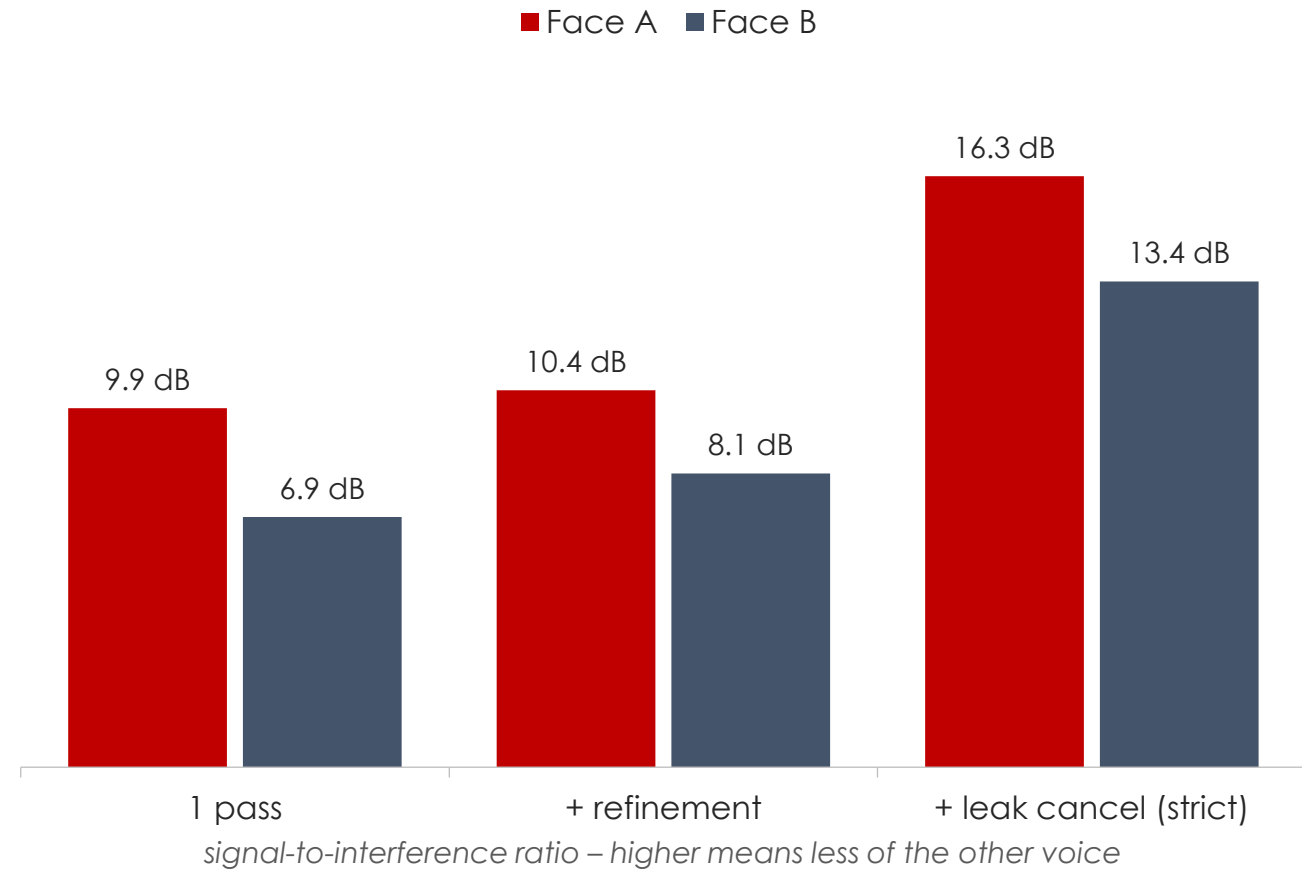


3.4 Design: Simulation

Ground-truth leak bench, built from our own recording

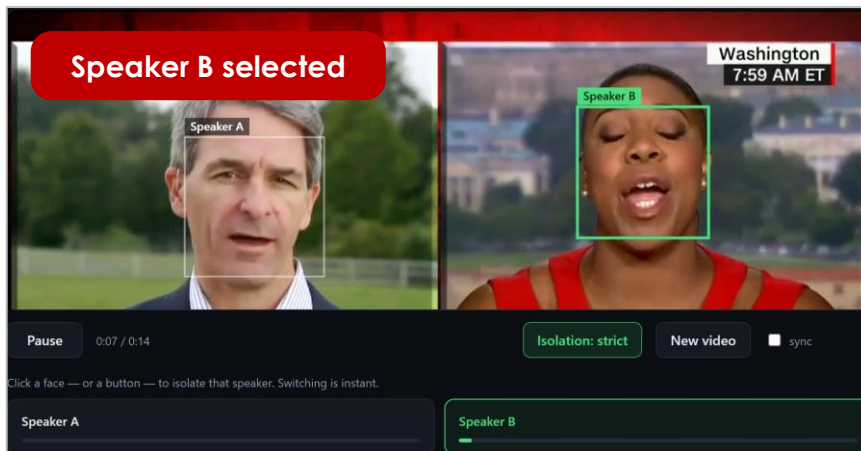
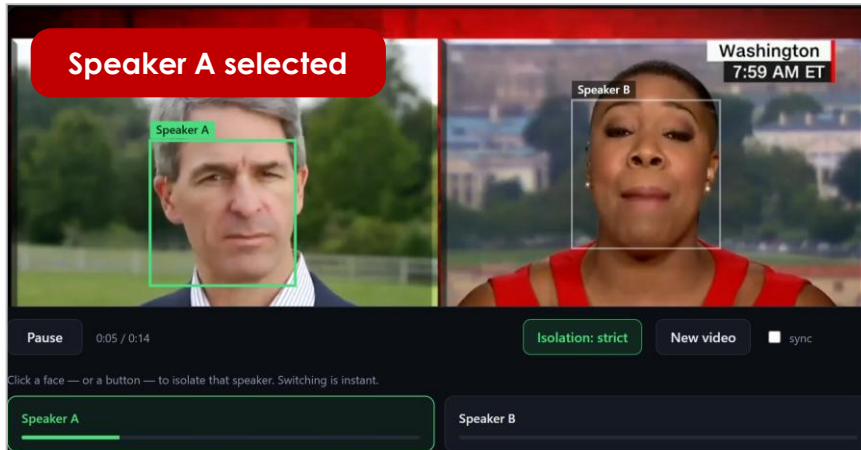


each source keeps its own lips; the 18 s shift makes the words unrelated



4 Implementation: Demonstration

Best of 5 test videos: split-screen debate, man and woman talking over each other



Face A lips	+0.78	+0.17	+0.39
Face B lips	+0.14	+0.32	+0.26
	Isolated channel A	Isolated channel B	Original mixture

Face × voice correlation

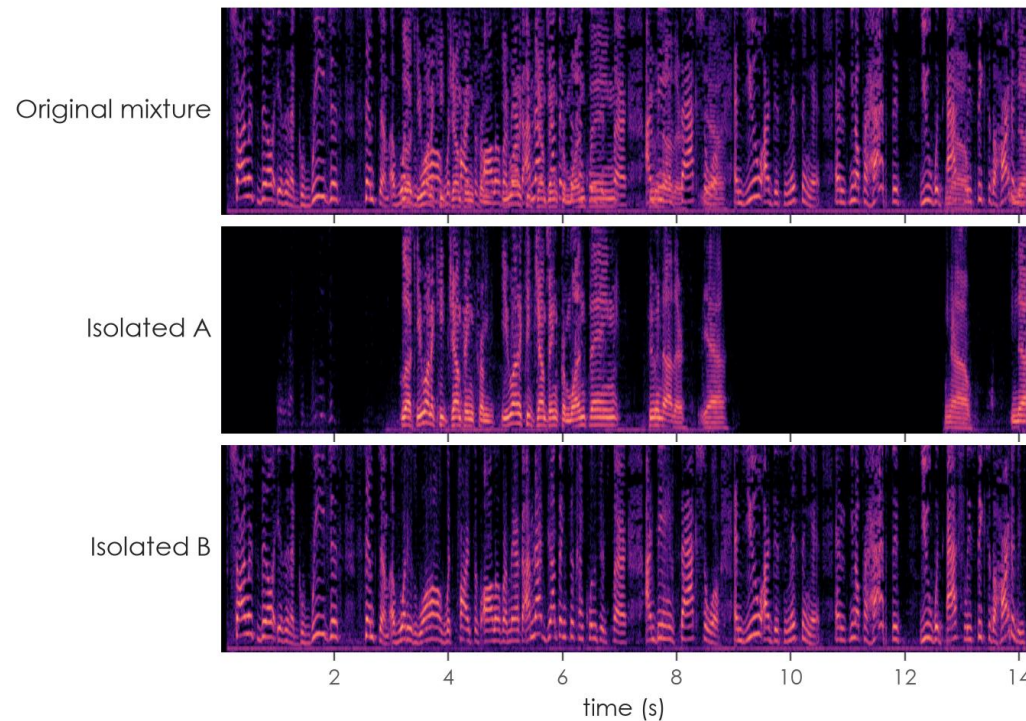
lip motion (mouth-image change) vs. channel loudness, Pearson r, 25 fps
diagonal wins: each face moves with its own isolated channel

0.08

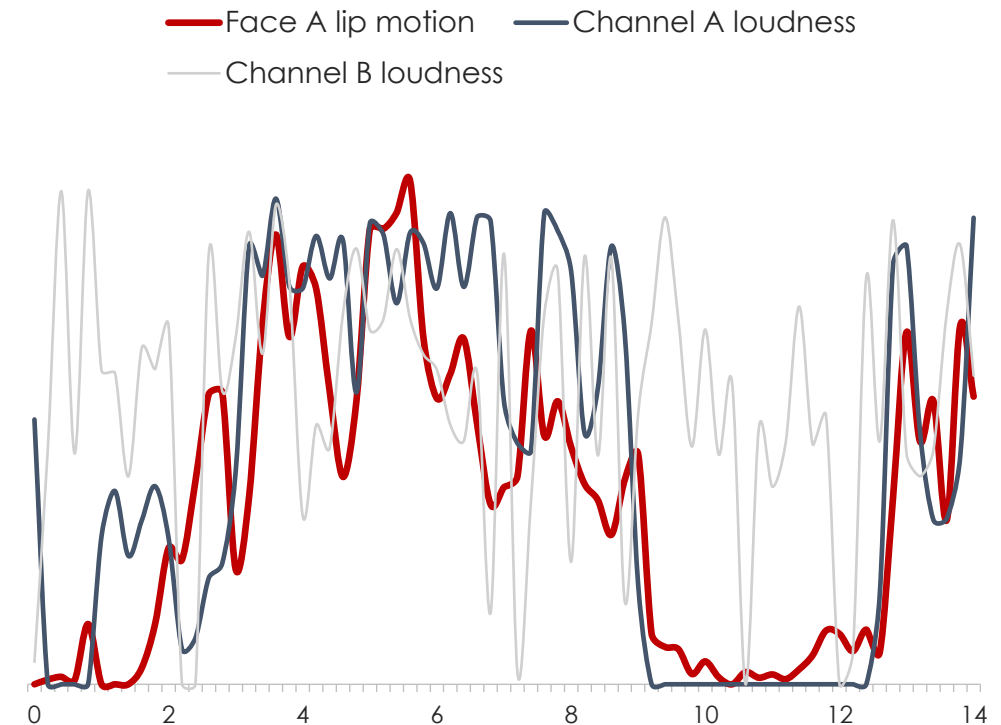
voice similarity between the two isolated channels
(0 = two different people, 1 = the same audio)



4 Implementation: Demonstration

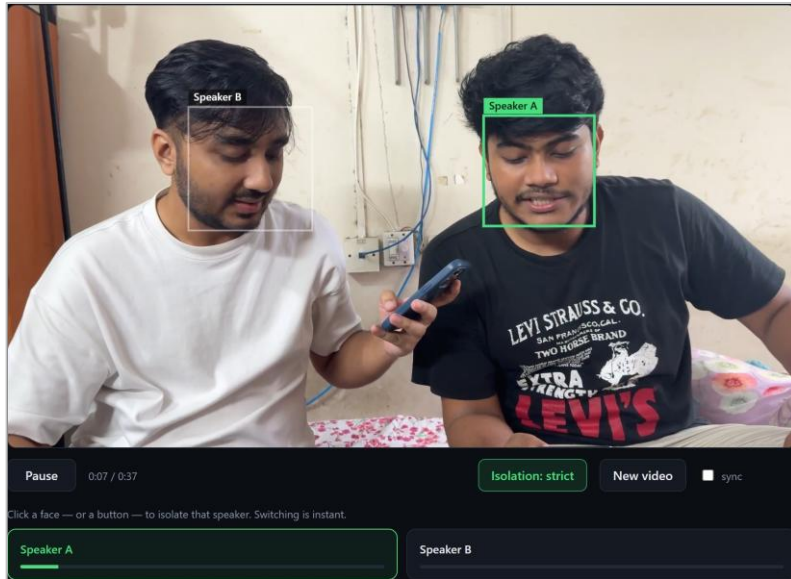


same 14 s, 0–4 kHz: channel A is silent between the man's interjections



*time (s) – the man's lips (red) rise and fall with his own channel (blue),
not the woman's (grey)*

4.1 Implementation: Photo Gallery



Our phone recording – Speaker A isolated



Bengali talk show with camera cuts – 3 people tracked



Re-identification across camera cuts: the same person keeps the same label (1 s, 11 s, 21 s, 32 s, 42 s)



5. Design Analysis and Evaluation

- 5.1 Novelty
- 5.2 Design Considerations (PO(c))
- 5.3 Investigations (PO(d))
- 5.4 Limitations of Tools (PO(e))
- 5.5 Impact Assessment (PO(f))
- 5.6 Sustainability and Environmental Impact Evaluation (PO(g))
- 5.7 Ethical Issues (PO(h))



5.1 Novelty

- 1 Lip-conditioned extraction replaced a chance-level matcher
- 2 Coherence-protected refinement: subtract others, never yourself
- 3 Leak canceller + live strict/natural switch in one buffer
- 4 Appearance re-ID across camera cuts; off-screen faces muted
- 5 Ground-truth leak bench built from our own recordings

Progress-update challenges → now

× Only 2 speakers

✓ **N faces, one extraction each**

× Voice shifting mid-clip

✓ **face-conditioned: no permutation**

× Leftover background

✓ **leak cancel + exact-zero gate**

× Lip-tracking accuracy

✓ **frame-aligned mouth ROI**

× Face leaving the frame

✓ **re-ID + auto-mute off screen**



5.2 Design Considerations (PO(c))

5.2.1 Public health & safety

- Runs locally – nothing uploaded (127.0.0.1)
- Click-free ramps, one global gain: no loudness jumps
- Exact digital silence in pauses

5.2.2 Environment

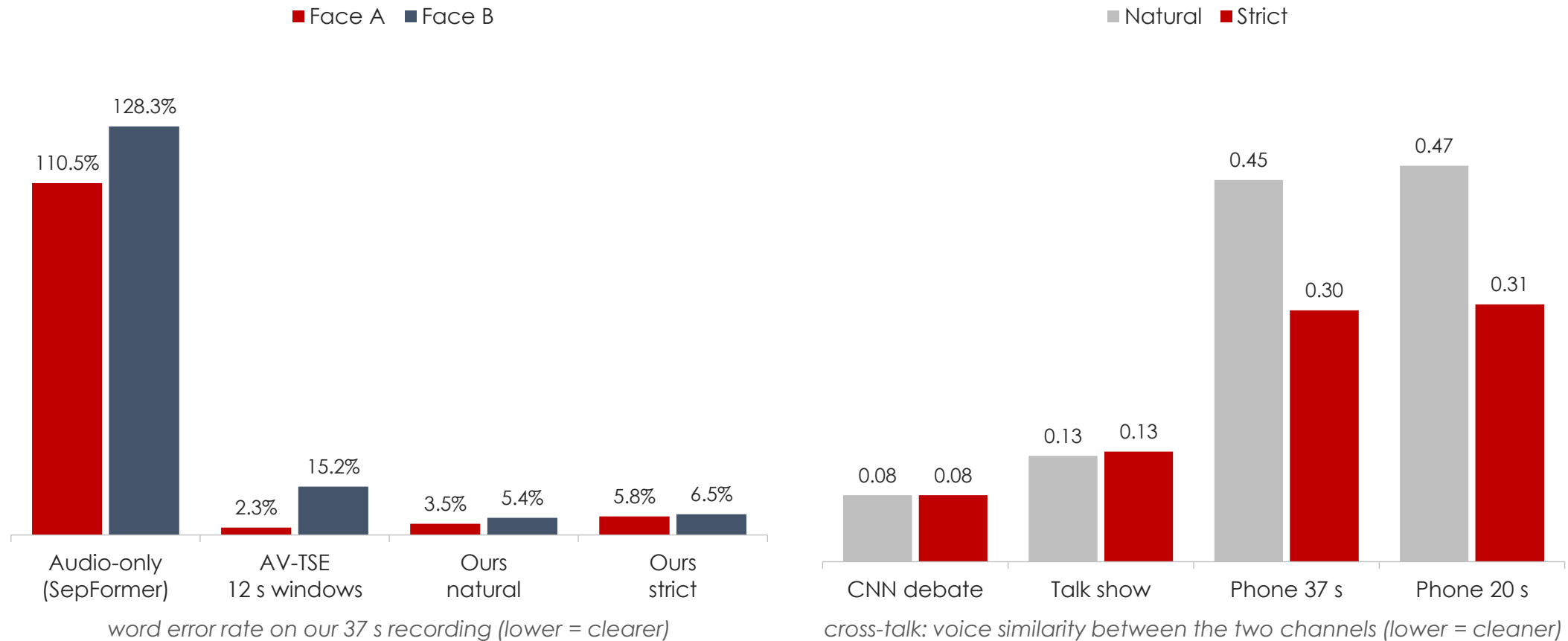
- One-time processing on a laptop GPU (~90 W)
- Skips off-screen windows: –52 % compute on edited video
- TF32: –17 % time, identical output

5.2.3 Cultural & societal needs

- Language-independent: tested in English and Bengali
- Helps hearing-impaired listeners follow one voice
- Tested man–woman and two similar male voices



5.3 Investigations (PO(d))



Tried and rejected: Dolphin (2026 model) · extra passes · sharp masks · speaker-ID gate – all measured worse



5.4 Limitations of Tools (PO(e))

AV-MossFormer2

16 kHz only → nothing above 8 kHz; 's' and 'f' sound duller

Lip video

faces must be visible, $\geq \sim 150$ px, 25 fps; off-screen voices can't be isolated

Similar voices, nonstop overlap

other voice still ~ 13 – 16 dB down in strict mode

RTX 5060 Laptop, 8 GB

$\sim 2\times$ real time; uploads capped at 5 min

MediaPipe 0.10.21

pins Python 3.12 and numpy < 2

Evaluation

proxies (Whisper WER, lip correlation) – no studio ground truth



5.5 Impact Assessment (PO(f))

5.5.1 Societal & cultural

- Accessibility for hearing-impaired listeners
- Meetings, classes, interviews, news archives

5.5.2 Health & safety

- Basis for hearing-assist tools
- No sudden loud artefacts; no physical risk

5.5.3 Legal

- Consent and privacy of recorded people
- Broadcast clips used only for testing
- Model weights: research licences (non-commercial)



5.6 Sustainability Evaluation (PO(g))

≈ 3 Wh

GPU energy per minute
of video (91 W cap ×
~2× real time)

0

new hardware – runs on
a laptop we already
own

0

training runs – open
pretrained models
reused

-52 %

compute on edited
video by skipping off-
screen faces

- Local processing avoids uploading video to data centres
- Open-source stack keeps the system maintainable and reusable



5.7 Ethical Issues (PO(h))

Consent

Our own recordings, with every member's consent; broadcast clips for testing only

Privacy

All processing stays on the machine; nothing is uploaded or stored elsewhere

Misuse

Needs a visible face; does not enhance voices of people who are off screen

Honesty

We report residual leakage, failures and rejected methods, with numbers

AI assistance

AI coding tools were used and are acknowledged



6. Reflection on Individual and Team work

6.1 Individual Contribution of Each Member

6.2 Mode of TeamWork

6.3 Diversity Statement of Team

6.4 Log Book of Project Implementation



6.1 Individual Contribution of Each Member



Redwan Nur Alam Shilad

2206177

- Literature review & problem scope
- Test-video set & success criteria
- Introduction & PO(b) sections

Presents slides 1–8



Ahnaf Hasan Adeeb

2206178

- System architecture & AV-TSE integration
- DSP: refinement, leak canceller, gate
- FastAPI backend & RTX 5060 setup

Presents slides 9–14



S. M. Khairul Islam Rifat

2206179

- Ground-truth leak bench & WER scoring
- Model comparisons: SepFormer, Dolphin
- Re-ID testing on edited video

Presents slides 15–22



Md. Shahriare Arefin Shuvo

2206180

- Vision setup: MediaPipe fix, face tracking
- Web Audio player & UI
- Ethics, cost analysis & docs

Presents slides 23–31



6.2 Mode of TeamWork and Diversity

How we worked

- One design document as the single source of truth (V2 $\rightarrow$ V3)
- Measure first: 40+ diagnostic scripts, every claim numbered
- Regression clips re-run after every change
- Each member presents one quarter: problem, design, evaluation, wrap-up

Team diversity

- **Complementary skills:** DSP theory, deep-learning deployment, web engineering, experimental evaluation
- **Builders and checkers:** two members built, two framed and measured – every claim was tested by someone else
- **Shared ownership:** each member presents one quarter and can explain the whole system
- **Inclusive testing:** English and Bengali, male and female voices, phone and broadcast video



6.3 Logbook of Project

Before Aug 10	Colab prototype: SepFormer + lip-correlation matching
Aug 10–11	V2 design; SepFormer + Wiener mask + Schmitt gate; FastAPI + Web Audio player
Aug 12	Diagnosis: lip–energy matcher no better than chance
Sep 7	Progress update presented
Sep 19–23	V3: AV-MossFormer2 integrated; identity gate; ghost test
Sep 24	RTX 5060; showcase recording; 2 s windows, refinement, leak cancel, re-ID, strict/natural UI

dates from the project's files and design documents



7 Communication to External Stakeholders (PO(j))

GitHub

[add repository link]

YouTube

[add demo video link]

Video title: Audio-Visual Speaker Isolation | C2.4 | EEE 312 (2026) Final Project | Dept of EEE, BUET

What we share: source code, README with setup steps, design documents (V2, V3), and ready-to-play demo jobs



8. Project Management and Cost Analysis (PO(k))

Item	Qty	Cost (BDT)
Laptop, RTX 5060 8 GB (already owned)	1	0
Smartphone camera (already owned)	1	0
Python, PyTorch, FastAPI, MediaPipe, SpeechBrain (open source)	–	0
Pretrained models: AV-MossFormer2, SFace, YuNet	–	0

0 BDT

extra cost per prototype

≈ 3 Wh

compute per minute of video
(mass deployment, software)



9. Future Work (PO(I))

1

Ground truth

record solo takes → SI-SDR, PESQ, STOI

2

Harder overlap

fine-tune on similar voices talking nonstop

3

Full bandwidth

restore the band above 8 kHz

4

Off-screen voices

voice enrollment when a face is not visible

5

Real time

streaming inference for live calls



10. References

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- [2] S. Zhao et al., "MossFormer2: combining Transformer and RNN-free recurrent network for enhanced time-domain monaural speech separation", ICASSP 2024.
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